

PERSONAL DATA



ASRATE GAULLE

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Date of Birth: 24/11/1990GC, Place of Birth: Alem Ketema, Shewa, Ethiopia

EDUCATION

- **Ph.D student** | *Computational Astrophysics* – |01/04/2025 - Present
 - Department of Physics, Ariel University, Ariel, 4070000, Israel
 - Astrophysics, Geophysics and Space Science (AGASS) Center, Ariel University, Ariel, 4070000, Israel
 - The proposed title of thesis: Radiative Transfer and Instability Evolution in Colliding Winds of Massive Binaries
 - Supervisor: Prof. Amit Kashi (Computational Astrophysicist)
- **Master of Science** | *Astronomy and Astrophysics with CGPA of 3.98/4.00* |05/10/2016 - 05/07/2018
 - Entoto Observatory & Research Center joint with Addis Ababa University, Addis Ababa, Ethiopia
 - Thesis title: An alternative method for measuring accretion rate in Galaxies using X/O flux ratio
- **Bachelor of Science** | *Major: Applied Physics with CGPA of 3.84/4.00* |05/10/2010 - 30/06/2013
 - Debre Berhan University, Debre Berhan, Ethiopia
 - Senior Project and Research Title: Optimizing Wind Energy Conversion to Electrical Power by calculating Betz-limit
- **General Science Stream** | *Natural Science* |05/10/2007 - 05/07/2010
 - Abregnoch Senior and Preparatory School, Alemketema, Shewa, Ethiopia

WORK EXPERIENCE

- *Current and Previous Employments*
 - Head, Department of Physics, 13/01/2021 - 24/06/2024, Dilla University (DU), Dilla, Ethiopia
 - Directed departmental academic and research activities.
 - Mentored students and faculty in curriculum development and research pursuits.
 - Chairperson of Ethiopian Space Science Society, 11/07/2020 -01/04/2025, DU Branch, Dilla, Ethiopia
 - Lecturer of Physics & Astrophysics, 06/07/2018 - 01/04/2025, Physics Department, DU, Dilla, Ethiopia
 - Taught undergraduate courses in physics, and developed interactive teaching materials for enhanced learning.
 - Assistant Lecturer of Physics, 20/09/2014 - 05/07/2018, Physics Department, DU, Dilla, Ethiopia
 - Graduate Assistant of Physics, 20/11/2013 - 20/09/2014, Physics, Haramaya University, Ethiopia
 - Vice President of Ethiopian Space Science Society, 22/10/2012 - 23/08/2013, Debre Birhan, Ethiopia

RESEARCH INTERESTS

- *My research interests (current & future) will be in Astrophysics (stellar,planetary, galactic, extragalactic) and Cosmology*
 - *Light Curve and Spectroscopic Analysis:* Examining the temporal and spectral properties of supernovae to characterize their physical and chemical properties.
 - *Dark Energy & Modified Gravity:* Studying $f(R, T, \&S)$ models and their impact on cosmic expansion, structure formation, and the relationship with dark energy and gravity theories.
 - *Cosmology:* Investigating LSS, galaxy formation-evolution, and the roles of DM and DE in cosmic expansion.
 - *Accretion & Feedback Processes:* Exploring the accretion mechanisms in galaxies and stars, along with the feedback effects from accretion disks, supernovae, and galaxy evolution.
 - *Massive Binaries:* Investigating the formation and evolution of massive binary systems, focusing on their accretion processes, interaction dynamics, and their role in feedback mechanisms.
 - *Galaxy Formation & Evolution:* Investigating the processes that drive the formation, growth, and evolution of galaxies, including the role of dark matter, star formation, gas dynamics, and feedback mechanisms.
 - *Planet Formation & Exoplanets:* Researching planetary formation, evolution, and exoplanet detection.
 - *Black Holes:* Investigating black hole formation, growth, accretion, mergers, and their role in galaxy evolution.

TEACHING EXPERIENCE AND SUPERVISION OF STUDENTS

- I have taught a wide range of courses and supervised BSc graduating senior projects in:

- Teaching courses in the Regular, Summer, STEM and Weekend programs, including as
 - General Physics, Mechanics, Electromagnetism, Experimental Physics in General Physics, Mechanics, Electromagnetism, Nuclear physics and Electronics
 - Astronomy (Fundamental and General), Introduction to: Astrophysics, Relativity and Cosmology
 - Computational Physics using Python and Introduction to Programming for Physics using FORTRAN
 - Research Methods & Senior project for Physicists, and Methods of Physics Teaching
- Supervision of students in the Regular, Summer, and Weekend programs,
 - I have assisted and supervised over one hundred Physics bachelor students in the Regular, Summer, and Weekend programs, advising senior projects across various areas of Physics, Astrophysics, Astronomy, and Space Science.

PROJECT AND RESEARCH

- An alternative method for measuring accretion rate in Galaxies | *Python & TOPCAT*|Spring 2018
 - Entoto Observatory and Research Center joint with Addis Ababa University
- AGN & Star-Formation Properties of Inside-out Assembled Galaxy| *Python & TOPCAT*|Spring 2018
 - Entoto Observatory and Research Center joint with Addis Ababa University
- Dark energy Models & its impact for expansion of Universe; and Massive Binaries | *Python*|Fall 2020
 - Dilla University, Dilla, Ethiopia

SCHOLARSHIP AWARDS

- | | |
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| • Grant of 6000 Shekel from Ariel University for Ph.D. studies in Israel (\$1820/month) | March 2025 - Present |
| • Grant of \$9500 from the Ministry of Education for M.Sc. studies | Sep 2016 – Jul 2018 |
| • Grant of \$2000 from the Ministry of Education for B.Sc. studies | Oct 2010 – Jul 2013 |

CONFERENCE AND PRESENTATIONS

- Grant to attend 17th International Conference on DARK SIDE OF THE UNIVERSE| 10 - 14 July 2023
 - ICTP-EAIFR, Kigali, Rwanda, <https://indico.ictp.it/event/10197/session/1/contribution/22>
- Grant to attend the International Astronomical Union (IAU) symposium 356|October 7th to 11th, 2019
 - International Astronomical Union (IAU) symposium 356
- Grant to attend the Panchromatic Universe conference organized by ESSTI-EORC|July 10th to 19th, 2017
 - Ethiopian Space Science and Technology Institute
- Grant to attend the fourth Middle East and African IAU Regional meeting (MEARIM)|May 22nd to 25th, 2017
 - East African Regional Office of Astronomy for Development
- Grant to attend the 39th International School for Young Astronomers (ISYA)|May 8th to 19th, 2017
 - International Astronomical Union
- Virtual Annual Conference of African Astronomical Society (AfAS2021)|March 8 - 12, 2021
 - African Astronomical Society¹

SYMPOSIUM, WORKSHOP AND TRAINING SCHOOLS

- First Pan-African Astro-Particle and Collider Physics Workshop|March 21 - 23, 2022
 - Ethiopian Space Science and Technology Institute
- National Ethiopian Workshop in Observational Astronomy and Data Reduction|March 3 - 5, 2022
 - Ethiopian Space Science and Technology Institute
- Online Workshop MW-Gaia: Bringing the Milky Way to Schools|June 2 - 4, 2021
 - Instituto de Astrofísica e Ciências do Espaço (Porto) and Vilnius University
- IAU Symposium 356: Nuclear Activity in Galaxies Across Cosmic Time|October 7 - 11, 2019

¹<https://www.africanastronomicalsociety.org/afas2021-posters/>

- *International Astronomical Union*
- Training School on Python Programming and Writing Skills|October 5 - 6, 2019
 - *IAU Symposium 356, Addis Ababa, Ethiopia*
- European Week of Astronomy and Space Science|April 3 - 6, 2018
 - Liverpool, UK *Presented by Dr. Mirjana Povic*

GIVEN TRAINING AND COMMUNITY SERVICE

- Training Physics Virtual Lab for Physics Teachers (PhET, VPlab, Stellarium) |01-06/12/2024
 - *Provided Hand On Experience in Exploring the basic concept of Physics and astronomy for Physics Teachers*
- Deraro Teaching and Tutoring|12/01/2019-30/06/2019
 - *Provided tutoring services to students in Physics*
- Hostel Tutorial Program for Top three Students from different schools|09/11/2023-30/06/2023
 - *Organized a tutorial program aimed at enhancing academic performance*
- Python Training for College Staff|24/09/2022-28/09/2022
 - *Conducted training sessions to improve programming skills among staff*
- Giving inspirational Lectures on Astronomy/Astrophysics, Cosmology, Atmospheric physics, Computational Astrophysics using python & Space Physics for STEM students|Every Year Since 2019-present
 - *Delivered lectures for special STEM students and the broader community*
- Hase Haro Special Tutorial for Grade 12 Students|12/01/2024-30/06/2024
 - *Provided targeted tutoring for grade 12 students preparing for exams*
- Work as Demonstrator of Physics Laboratory for University Community during science exhibition day when I was BSc student, Debre Berhan University, Ethiopia |March 30 - 31, 2013
 - *Assisted in demonstrating physics experiments during the university's science exhibition day as a BSc student.*

COMPUTATIONAL AND PROGRAMMING SKILLS

- Professional Skills such as:
 - *Languages:* English (Excellent and Fluent), Amharic (Native)
 - *Programming:* Python ([NumPy](#), [SciPy](#), [Matplotlib](#), [SunPy](#), [AstroPy](#), [Pandas](#), [MagPy](#), [PHOEBE](#), [YT](#), [FLASH](#), [RADMC](#), [TORUS](#), [SymPy](#), [Seaborn](#), [PyRaf](#), [Pylab](#)), [R](#), [AstroImageJ](#), TOPCAT, IRAF, DS9, C++, pynbody and FORTRAN
 - *Document Creation:* Professional [L^AT_EX](#), Microsoft-Office-Suite and WPS-Writer (word, ppt, excel) user
 - *Excellent written and verbal communication skills*
 - *Analyze large datasets using advanced software and statistical methods*

PUBLICATIONS

- AGN and Star-Formation Properties of Inside-out Assembled Galaxy Candidates at $z < 0.1$ |2020
 - Zewdie. D., Pović. M., Aravena. M., Roberto J., and Gaulle. A.|MNRAS, 498, 4345
- Properties of local galaxies with the highest growth rate at redshift $z \leq 0.1$ |2019
 - Zewdie. D., Pović. M., Aravena. M., Roberto J., and Gaulle. A.|Proceedings IAU Sm No.356,arXiv:2004.00718
- Accretion rate in AGN and X-ray-to-optical flux ratio at $z < 0.2$ |2019
 - Gaulle. A., Pović. M., and Zewdie. D.|Proceedings IAU Sym No.356,arXiv:2003.13487
- An Alternative Method for Measuring the Accretion Rate in Active Galactic Nuclei at $z \leq 0.2$ |2025
 - Gaulle. A., Pović. M., and Zewdie. D.|A&A, to be submitted

ACADEMIC REFERENCES

- Prof. Mirjana Povic|PhD in Astrophysics (Advisor)
 - Ethiopian Space Science and Technology Institute Addis Ababa, Ethiopia|E-mail: mpovic@iaa.es
- Dr. Getinet Feleke|PhD in Astrophysics
 - Kotebe Metropolitan University, Addis Ababa, Ethiopia| E-mail: getinetfelek@gmail.com
- Dr. Remudin Reshid|PhD in Astrophysics
 - Addis Ababa University, Addis Ababa, Ethiopia| E-mail:remudinreshid@gmail.com